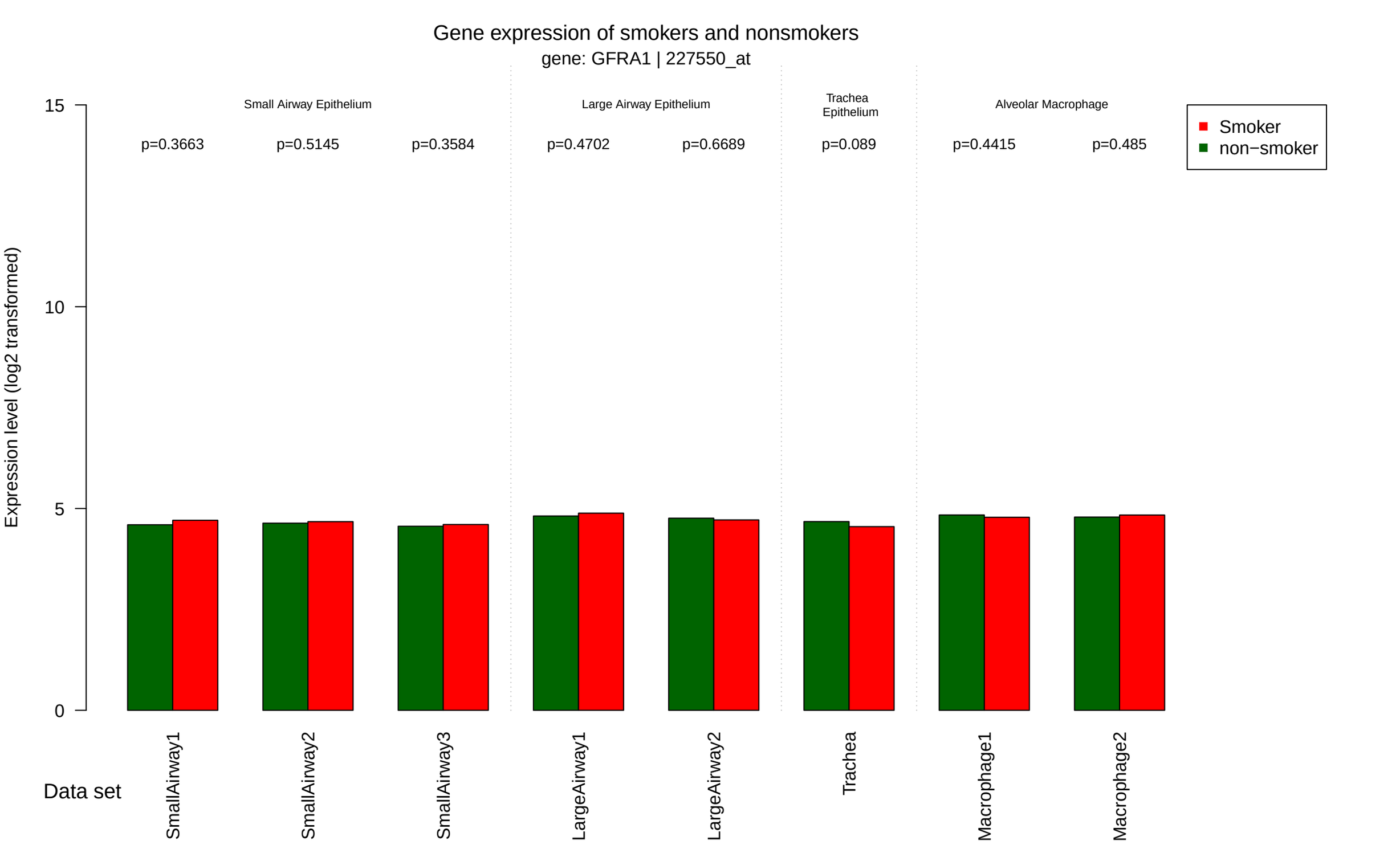
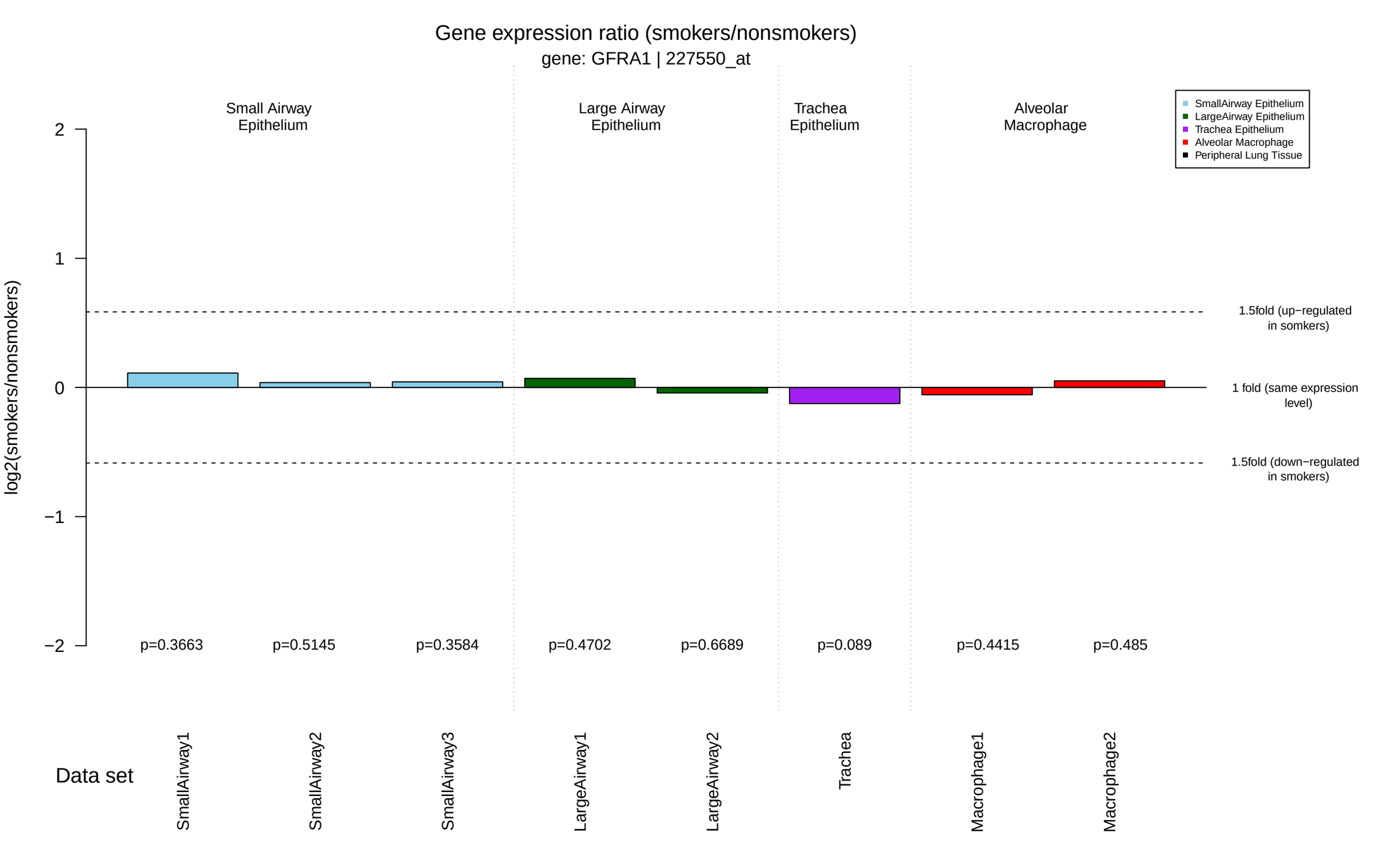
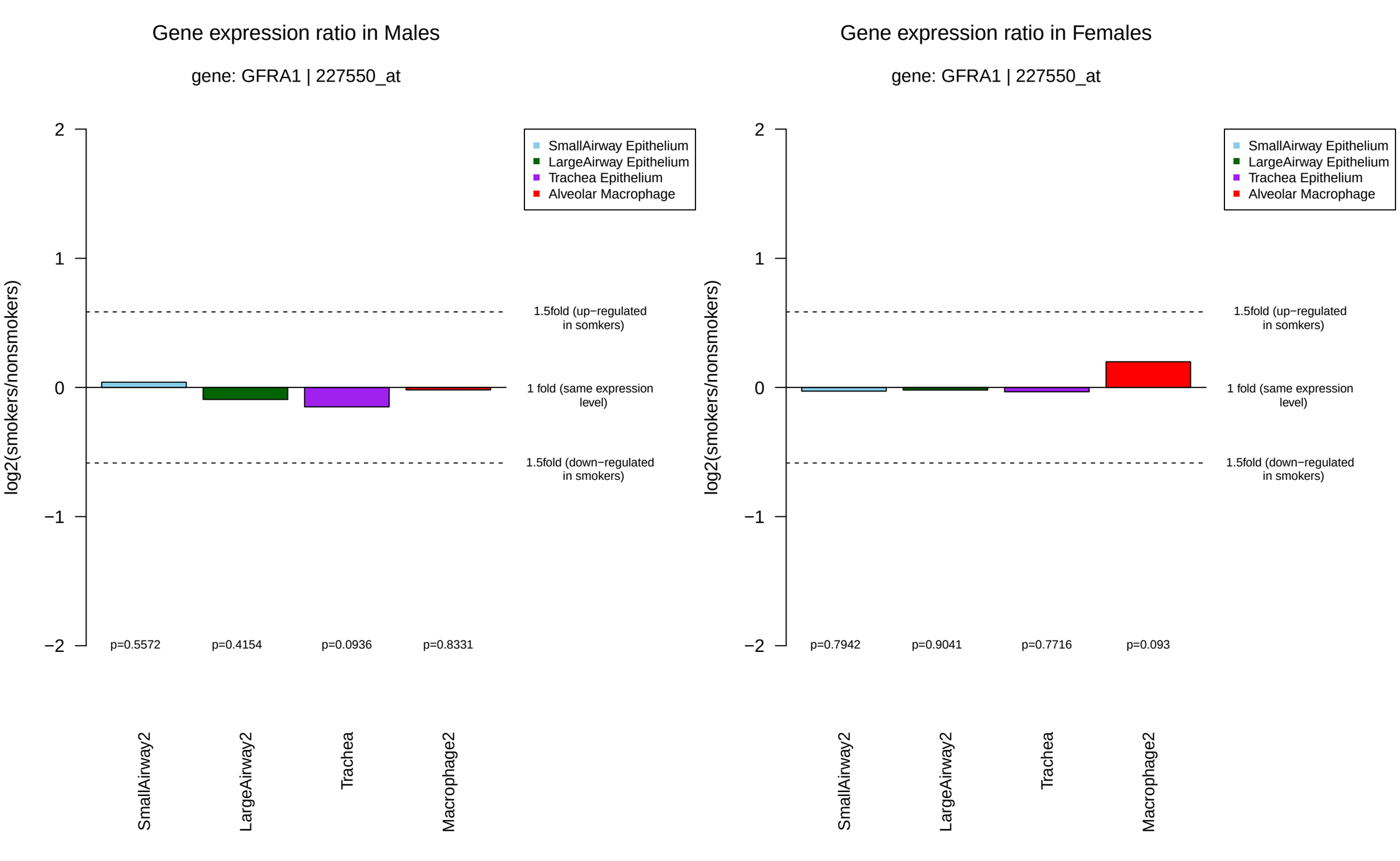
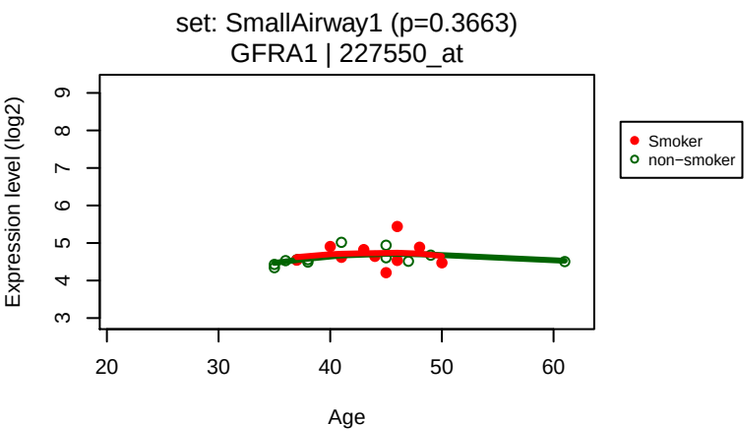
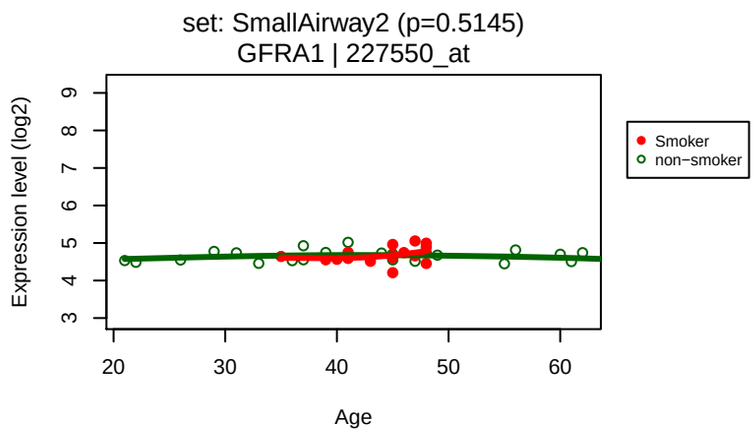
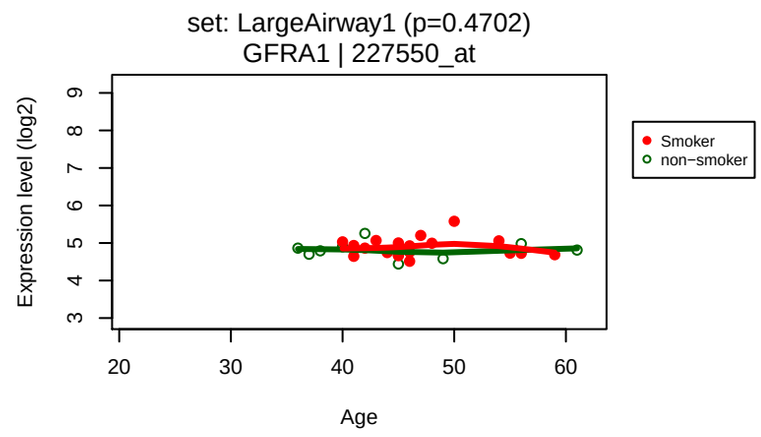

Gene expression of smokers and nonsmokers
gene: GFRA1 | 227550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3663
p=0.5145
p=0.3584
p=0.4702
p=0.6689
p=0.089
p=0.4415
p=0.485
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

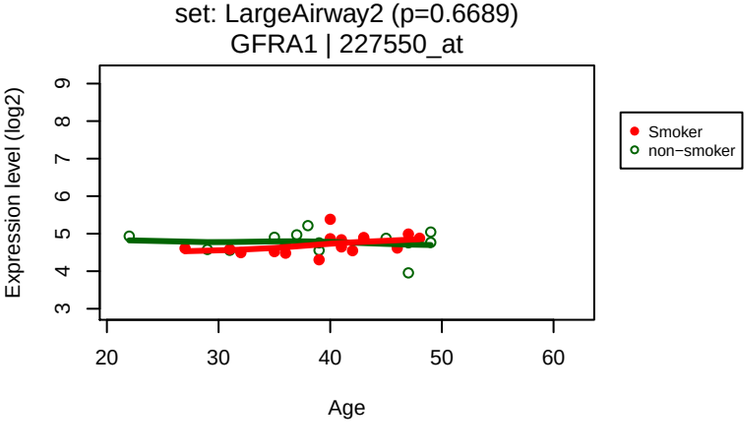

Gene expression ratio (smokers/nonsmokers)
gene: GFRA1 | 227550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3663
p=0.5145
p=0.3584
p=0.4702
p=0.6689
p=0.089
p=0.4415
p=0.485
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

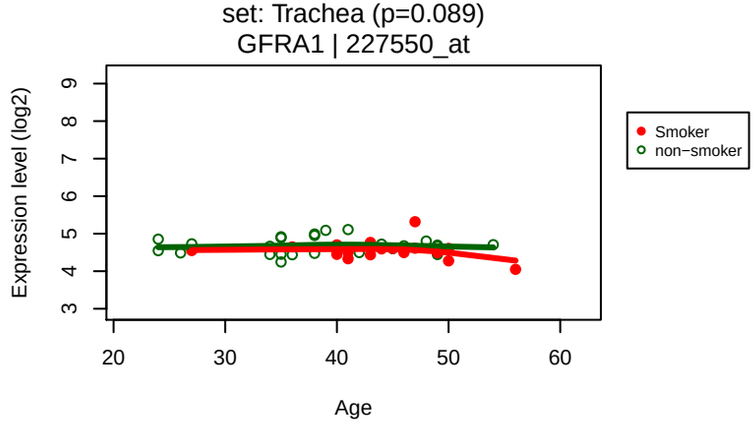

Gene expression ratio in Males
gene: GFRA1 | 227550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5572
p=0.4154
p=0.0936
p=0.8331
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GFRA1 | 227550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7942
p=0.9041
p=0.7716
p=0.093
SmallAirway2
LargeAirway2
Trachea
Macrophage2

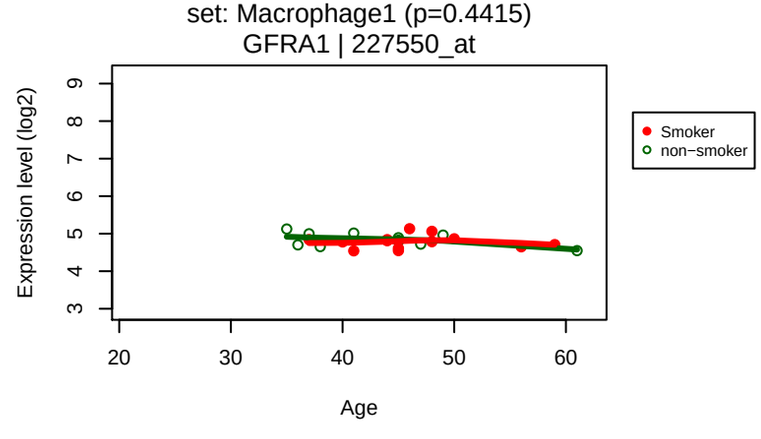

set: SmallAirway1 (p=0.3663)
GFRA1 | 227550_at
Expression level (log2)
Age
Smoker
non−smoker

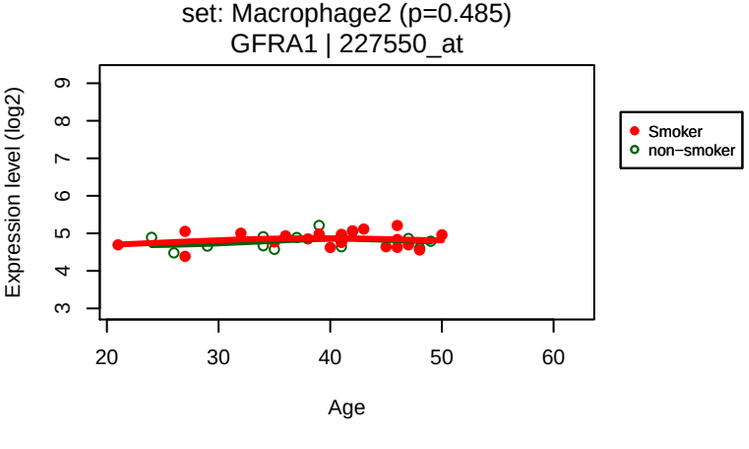

set: SmallAirway2 (p=0.5145)
GFRA1 | 227550_at
Expression level (log2)
Age
Smoker
non−smoker

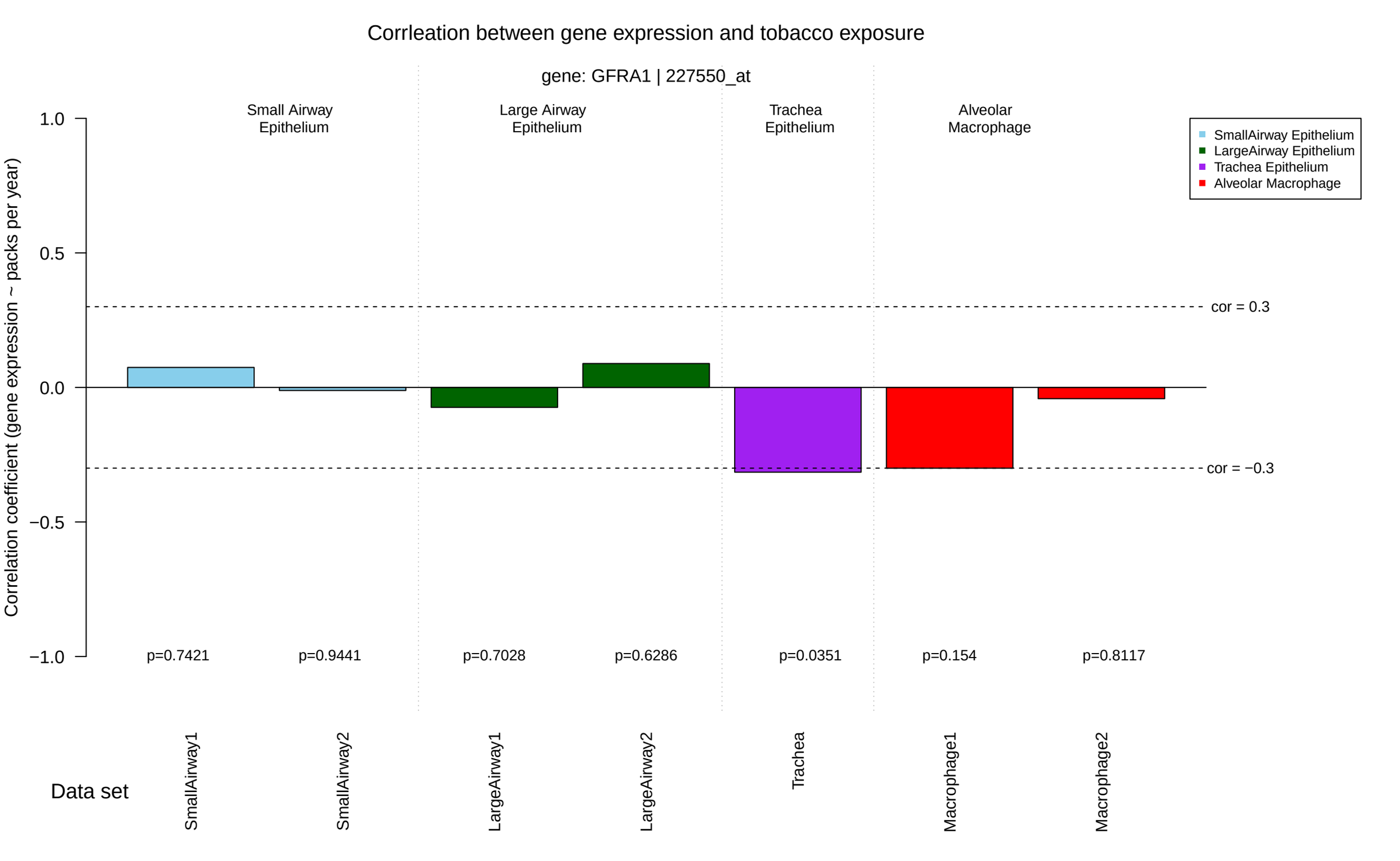

set: LargeAirway1 (p=0.4702)
GFRA1 | 227550_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6689)
GFRA1 | 227550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.089)
GFRA1 | 227550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4415)
GFRA1 | 227550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.485)
GFRA1 | 227550_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GFRA1 | 227550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7421
p=0.9441
p=0.7028
p=0.6286
p=0.0351
p=0.154
p=0.8117
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2